
CSE (Data Science)

Major Project : Phase-I

A.Y : 2026-27

Year & Sem : IV & I Sem

Date : 07.07.2025

Title of the project:

VisionForge AI: An Explainable Vision-Language Framework for Industrial Defect Detection, Root Cause Analysis, and Automated Inspection Reporting

Project Batch No:

S.No.	H.T.No.	Name	Mail id (personal)	Mobile No.	Sign
1	23K91A6755	G.SRIVARDHAN	srivardhangandla858@gmail.com	6309129611	
2	23K91A6733	B.TRIVED	trivedbobbali@gmail.com	9848143903	
3	23K91A6752	G.RAKESH	gaddalarakesh18@gmail.com	9177136935	
4	23K91A6729	B.INDRA NEEL	indraneelvarma120@gmail.com	8519988616	

Guide Name : **Mr K.SATHYANARAYANA**

Designation : **Assistant Professor**

Mobile No : **9963282737**

Mail id : **satyam@tkrcet.com**

Abstract:

VisionForge AI is an intelligent software-based industrial inspection system designed to improve product quality by automating the defect detection process. The system uses deep learning-based computer vision techniques to identify defects in industrial products such as Printed Circuit Boards (PCBs), steel surfaces, and other manufactured components. Unlike traditional AI models that only classify defects, VisionForge AI also explains why a defect was detected by highlighting the important regions of the image using Explainable AI techniques.

The system further enhances the inspection process by generating a simple human-readable explanation and suggesting possible root causes for the detected defect. A web-based dashboard allows users to upload images, view inspection results, analyze explanation heatmaps, and automatically generate professional inspection reports. The reports include defect type, confidence score, explanation, possible causes, and recommended corrective actions.

The proposed system is completely software-based, making it cost-effective and easy to implement using publicly available industrial datasets. VisionForge AI aims to improve inspection accuracy, reduce manual effort, increase transparency in AI predictions, and support faster decision-making in manufacturing industries.

Project Coordinator

Project Guide